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Research Interest

Deep Learning, Training Dynamics, Statistical Learning Theory, Foundation Models, Reinforcement Learning

Research Highlights

Thread I: Foundations of Learning, Training Dynamics & Interpretability

How transformers pick up features and circuits during training, and when learning is possible at all.

- **Taming Polysemanticity in LLMs: Theory-Grounded Feature Recovery via Sparse Autoencoders**
  Gives the first provable guarantees for feature recovery with sparse autoencoders. The resulting Group-wise Bias Adaptation algorithm finds cleaner, more interpretable features in real LLM activations.
  ICLR 2026 · **Invited Talk**, FAI Seminar 2025
- **Unveiling Induction Heads: Provable Training Dynamics and Feature Learning in Transformers**
  First end-to-end characterization of how a transformer with an FFN learns induction heads, the attention circuits behind in-context learning.
  NeurIPS 2024 · **Invited Talk**, FAI Seminar 2024
- **Training Dynamics of Multi-Head Softmax Attention for In-Context Learning: Emergence, Convergence, and Optimality**
  First end-to-end characterization of the full training dynamics of multi-head softmax attention, proving convergence to the optimal in-context learning algorithm.
  COLT 2024 (**Oral**) · ICLR 2024 Workshop on Bridging the Gap Between Practice and Theory in Deep Learning · **Invited Talks**, Workshop on the Theory of Decision Intelligence, RLChina 2024, and FAI Seminar 2024
- **In-Context Linear Regression Demystified: Training Dynamics and Mechanistic Interpretability of Multi-Head Softmax Attention**
  Proves that trained multi-head softmax attention runs one step of gradient descent on in-context linear regression.
  ICML 2025
- **How Transformers Learn Causal Structures In-Context: Explainable Mechanism Meets Theoretical Guarantee**
  Shows how transformers learn causal graph structure in context, with an explainable mechanism and theoretical guarantees.
  ICLR 2026
- **Can Neural Networks Achieve Optimal Computational-Statistical Tradeoff? An Analysis on Single-Index Model**
  Proves that neural networks trained by gradient descent reach the optimal sample complexity for single-index models, matching statistical-query lower bounds.
  ICLR 2025 (**Oral**)
- **An Optimized Franz–Parisi Criterion and Its Equivalence with SQ Lower Bounds**
  Builds an optimized Franz–Parisi criterion from statistical physics and proves it is exactly equivalent to statistical-query lower bounds for high-dimensional inference.
  NeurIPS 2025 (**Oral**)

Thread II: Pretraining, Scaling & Latent Reasoning

How data, architecture and optimizer choices shape what a model learns, how that changes with compute, and how models can reason in latent space.

- **Truncate or Not? A Scaling Study of Looped Transformers with Truncated Backpropagation**
  Develops a scaling law for looped transformers that jointly accounts for model size, training tokens, and forward and backward loop counts. At matched compute, it shows when full backpropagation through fewer loops beats truncation.
  Preprint, 2026
- **Unlocking Out-of-Distribution Generalization in Transformers via Recursive Latent Space Reasoning**
  Introduces recursive latent-space reasoning, which lets transformers generalize to inputs far outside the training range. Probing shows what the trained looped transformer computes at each loop.
  Journal of Machine Learning Research (minor revision)
- **Towards Understanding Momentum Acceleration in River-Valley Loss Landscape**
  Explains why momentum speeds up training on the river-valley loss landscapes seen in LLM pretraining: it allows a much larger learning rate, which drives faster progress along the river.
  NeurIPS 2026 · ICML 2026 Workshop on High Dimensional Learning Dynamics: the Science of Scaling (**Spotlight**)

Thread III: Post-training & Self-Evolution

Helping models improve their own reasoning through RL and curricula they write themselves.

- **INFUSER: Influence-Guided Self-Evolution Improves Reasoning**
  A generator and a solver train together. The generator writes problems grounded in documents and learns which ones actually help the solver, using influence scores. The solver's reasoning improves as the curriculum evolves.
  NeurIPS 2026 · **Invited Talk**, NICETalk 2026
- **GPT-Driven Drug Optimization with Structured Policy Optimization Post-training**
  Post-trains a GPT model with structured policy optimization to improve drug candidates, tested on SARS-CoV-2 and human cancer benchmarks.
  MLHC 2026
- **Benchmarking AI Agents for Addressing Scientific Challenges Across Scales**
  Introduces SciAgentArena, about 200 real research tasks with stepwise checks and an interactive environment for testing AI agents. Agents do well on clear data-analysis workflows but struggle with open-ended research.
  Preprint, 2026

Education

Doctoral Student — **Yale University**, Statistics and Data Science — Sept. 2022 to now

- *Coursework:* Linear models, Statistical Inference, Practical Work, Data Analysis, Intermediate Machine Learning, Advanced Optimization
- *Advisor:* Prof. Zhuoran Yang

Bachelor of Science — **Tsinghua University**, Electronic Engineering — Sept. 2018 to May 2022

- *Coursework:* Convex Optimization, Computer Program Design, Linear Algebra, Probability and Stochastic Processes, Theory and Methods for Statistical Inference
- *Honors & Awards:*
  - Comprehensive Excellence Scholarship
  - China National Scholarship

Experience

Yale University, Teaching Assistant — CT, USA

- Information Theory (S&DS 364) — Feb 2024 to May 2024
- Optimization and Computation (S&DS 431) — Sep 2024 to Dec 2024
- Deep Learning for Scientists and Engineers (S&DS 266) — Feb 2025 to May 2025

Google Research, Student Researcher — CT, USA
June 2025 to Dec 2025

- Research on test-time scaling behavior and workflow optimization for large language models.

Tencent Hunyuan, Qingyun intern — Beijing, China
Aug 2026 to now

- Omni LLM pretraining and gradient-information-based data improvement.

Publications

Chen, S., M. Lu, B. Wu, H. Sheen, F. Zhang, S. Li, Z. Li, J. Blanchet, T. Wang, and Z. Yang (2026). "INFUSER: Influence-Guided Self-Evolution Improves Reasoning". In: *The Fortieth Annual Conference on Neural Information Processing Systems*.

Chen, S., T. Misiakiewicz, I. Zadik, and P. Zhang (2026). "An optimized franz-parisi criterion and its equivalence with sq lower bounds". In: *Advances in Neural Information Processing Systems* 38. **Oral**, pp. 110461–110506.

Chen, S., H. Sheen, X. Xiong, T. Wang, and Z. Yang (2026). "Taming Polysemanticity in LLMs: Theory-Grounded Feature Recovery via Sparse Autoencoders". In: *The Fourteenth International Conference on Learning Representations*.

He, J., L. Wang, S. Chen, and Z. Yang (2026). "On the Mechanism and Dynamics of Modular Addition: Fourier Features, Lottery Ticket, and Grokking". In: *arXiv preprint arXiv:2602.16849*.

He, J., L. Wang, F. Zhang, S. Chen, and Z. Yang (2026). "Neural Networks Provably Learn Spectral Representations for Group Composition". In: *arXiv preprint arXiv:2606.02993*.

Hu, X., F. Zhang, S. Chen, and Z. Yang (2026). "Unveiling the Statistical Foundations of Chain-of-Thought Prompting Methods". In: *Journal of Machine Learning Research* 27.202, pp. 1–104. URL: https://www.jmlr.org/papers/v27/25-2240.html.

Liu, T., A. X. Wang, A. Panescu, L. X. Chen, W. Long, X. Wei, Y. Jing, Z. Zeng, J. Chen, S. Jiang, Z. Wang, S. Gu, S. Chen, X. Hu, H. Shao, L. Xu, W. Zheng, Z. Cao, A. Fang, B. Yu, K. Sun, R. Ying, A. Cohan, Q. Chen, L. Xue, K. Ding, Y. Du, W. Jin, Z. Yang, M. Zitnik, J. Zou, H. Xu, and H. Zhao (2026). "Benchmarking AI Agents for Addressing Scientific Challenges Across Scales". In: *arXiv preprint arXiv:2606.12736*.

Liu, X., S. Jiang, S. Chen, Z. Yang, Y. Chen, I. T. Foster, and R. L. Stevens (2026). "GPT-Driven Drug Optimization with Structured Policy Optimization Post-training". In: *Proceedings of the 11th Machine Learning for Healthcare Conference*. Vol. 340. Proceedings of Machine Learning Research, pp. 1208–1242.

Lu, M., Z. Bian, K. Wen, B. Wu, S. Chen, T. Wang, and Z. Li (2026). "Towards Understanding Momentum Acceleration in River-Valley Loss Landscape". In: *The Fortieth Annual Conference on Neural Information Processing Systems*.

Wei, J., S. Chen, J. He, and Z. Yang (2026). "How Transformers Learn Causal Structures In-Context: Explainable Mechanism Meets Theoretical Guarantee". In: *The Fourteenth International Conference on Learning Representations*.

Xiong*, X., S. Chen*, Z. Shao, Y. Su, W. Jin, P. Luo, Z. Yang, and X. Liu (2026). "Truncate or Not? A Scaling Study of Looped Transformers with Truncated Backpropagation". In: *arXiv preprint arXiv:2610.26030*. URL: https://arxiv.org/abs/2610.26030v2.

Altabaa, A., S. Chen, J. Lafferty, and Z. Yang (2025). "Unlocking Out-of-Distribution Generalization in Transformers via Recursive Latent Space Reasoning". In: *arXiv preprint arXiv:2510.14095*.

Chen, S., B. Wu, M. Lu, Z. Yang, and T. Wang (2025). "Can neural networks achieve optimal computational-statistical tradeoff? an analysis on single-index model". In: *International Conference on Learning Representations*. Vol. 2025. **Oral**, pp. 71176–71260.

Chen, Z., S. Chen, Z. Qi, X. Chen, and Z. Yang (2025). "Quantile-Optimal Policy Learning under Unmeasured Confounding". In: *arXiv preprint arXiv:2506.07140*.

He, J., X. Pan, S. Chen, and Z. Yang (2025). "In-Context Linear Regression Demystified: Training Dynamics and Mechanistic Interpretability of Multi-Head Softmax Attention". In: *International Conference on Machine Learning*. PMLR, pp. 22686–22742.

Liu, X., H. T. Le, S. Chen, R. Stevens, Z. Yang, M. R. Walter, and Y. Chen (2025). "Active Advantage-Aligned Online Reinforcement Learning with Offline Data". In: *arXiv preprint arXiv:2502.07937*.

Chen, S., H. Sheen, T. Wang, and Z. Yang (2024a). "Training Dynamics of Multi-Head Softmax Attention for In-Context Learning: Emergence, Convergence, and Optimality". In: *The Thirty Seventh Annual Conference on Learning Theory*. **Oral**. PMLR, pp. 4573–4573.

Chen, S., H. Sheen, T. Wang, and Z. Yang (2024b). "Unveiling Induction Heads: Provable Training Dynamics and Feature Learning in Transformers". In: *Advances in Neural Information Processing Systems*. Vol. 37, pp. 66479–66567.

He, J., S. Chen, F. Zhang, and Z. Yang (2024). "From Words to Actions: Unveiling the Theoretical Underpinnings of LLM-Driven Autonomous Systems". In: *International Conference on Machine Learning*. PMLR, pp. 17807–17841.

Sheen, H., S. Chen, T. Wang, and H. H. Zhou (2024). "Implicit Regularization of Gradient Flow on One-Layer Softmax Attention". In: *arXiv preprint arXiv:2403.08699*.

Wu, J., S. Chen, M. Wang, H. Wang, and H. Xu (2024). "Contractual Reinforcement Learning: Pulling Arms with Invisible Hands". In: *arXiv preprint arXiv:2407.01458*.

Chen, S., M. Wang, and Z. Yang (2023). "Actions Speak What You Want: Provably Sample-Efficient Reinforcement Learning of the Quantal Stackelberg Equilibrium from Strategic Feedbacks". In: *arXiv preprint arXiv:2307.14085*.

Chen, S., Y. Wang, Z. Wang, and Z. Yang (2023). "A unified framework of policy learning for contextual bandit with confounding bias and missing observations". In: *arXiv preprint arXiv:2303.11187*.

Chen, S., J. Wu, Y. Wu, and Z. Yang (2023). "Learning to incentivize information acquisition: Proper scoring rules meet principal-agent model". In: *International Conference on Machine Learning*. PMLR, pp. 5194–5218.

Chen, S., D. Yang, J. Li, S. Wang, Z. Yang, and Z. Wang (2022). "Adaptive model design for Markov decision process". In: *International Conference on Machine Learning*. PMLR, pp. 3679–3700.

Zhang*, Y., S. Chen*, Z. Yang, M. Jordan, and Z. Wang (2021). "Wasserstein flow meets replicator dynamics: A mean-field analysis of representation learning in actor-critic". In: *Advances in Neural Information Processing Systems* 34, pp. 15993–16006.

Technologies

Languages: Python, C/C++, R

LLM Development: PyTorch, Hugging Face Transformers, vLLM/sglang, Megatron, slime, verl

Training & Evaluation: pretraining, RLVR, deep research, benchmark evaluation